

FE-CH-USSD-01 - USSD Channel - Developer Implementation Guide

Version: v1.0 Bundle: 99_frontend Companion DD: FE-CH-USSD-01-USSD_Channel-v1.0

1. Goal

Implement USSD as a stateless gateway-facing adapter with server-side session state.

USSD must call owning module APIs. It must not implement local billing, payment, subscription, ticket, or customer state changes.

2. Runtime Flow

```
sequenceDiagram
    participant GW as USSD Gateway
    participant USSD as USSD Adapter
    participant ILM as ILM
    participant BIL as Billing
    participant SUB as SUB-LM
    participant TCK as TCK
    participant INT as CUST-INT

    GW->>USSD: POST session request
    USSD->>USSD: load/create session
    USSD->>ILM: resolve MSISDN
    USSD->>BIL: query balance if menu selected
    USSD->>SUB: query subscription if selected
    USSD->>TCK: create ticket if selected
    USSD->>INT: record interaction
    USSD-->>GW: CON or END text
```

3. Step 1 - Migrations

Create:

- ussd_session
- ussd_menu_definition
- ussd_request_log

Seed WIK English menus:

- MAIN
- BALANCE
- PAY
- SUBSCRIPTION
- SUPPORT
- LANGUAGE
- UNKNOWN_CUSTOMER

4. Step 2 - Gateway Endpoint

Endpoint:

```
POST /api/channels/ussd/sessions
Content-Type: application/json
```

Request:

```
{
  "operatorCode": "WIK",
  "gateway": "SAFARICOM_AGGREGATOR",
  "gatewaySessionId": "gw-abc-001",
  "gatewayRequestId": "gw-msg-001",
  "msisdn": "254700000001",
  "inputText": "1",
  "shortCode": "*123#",
  "requestedAt": "2026-08-30T08:01:00Z"
}
```

Implementation:

1. Normalize MSISDN.
2. Find or create ussd_session.
3. If gatewayRequestId already logged, return same response.
4. Resolve customer on first request:

```
GET /api/customers/lookup?msisdn=254700000001
```

5. Route input to next menu.
6. Log request/response.
7. Return CON or END.

5. Step 3 - Balance Menu

When customer selects balance:

```
GET /api/accounts/ACC-900045/balance
```

Optional dunning:

```
GET /api/dunning-states?accountId=ACC-900045
```

Render:

```
Balance KES 1200
Due 05 Sep
1 Pay now
2 Main menu
```

Keep text short and configurable.

6. Step 4 - Payment Menu

If payment instructions mode:

```
POST /api/payments/instructions
Idempotency-Key: ussd-gw-msg-001-payinst
Content-Type: application/json

{
  "operatorCode": "WIK",
  "accountId": "ACC-900045",
  "channel": "USSD",
  "msisdn": "254700000001"
}
```

If STK push is enabled, call PAY-GW-01 payment initiation API. Never mark payment successful from USSD callback alone; payment application belongs to BIL/PAY-GW flow.

7. Step 5 - Subscription Menu

Call:

```
GET /api/subscriptions?customerId=CUS-100045
```

Render only compact fields:

```
1 Fiber 50M ACTIVE
2 Main menu
```

If multiple subscriptions, let customer select one before showing details.

8. Step 6 - Support Menu

Create simple ticket:

```
POST /api/tickets
Idempotency-Key: ussd-gw-msg-001-ticket
Content-Type: application/json

{
  "operatorCode": "WIK",
  "ticketType": "USSD_SUPPORT_REQUEST",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "channel": "USSD",
  "summary": "Customer requested callback via USSD"
}
```

Return:

```
Request received.
Ref TCK-1001.
We will contact you.
```

9. Step 7 - Interaction Logging

After meaningful actions, call:

```
POST /api/customer-interactions
Idempotency-Key: ussd-gw-msg-001-interaction
Content-Type: application/json

{
  "operatorCode": "WIK",
  "customerId": "CUS-100045",
  "channel": "USSD",
  "interactionType": "USSD_BALANCE_VIEWED",
  "originRefType": "USSD_SESSION",
  "originRefId": "ussd-sess-001"
}
```

10. Step 8 - Timeout Worker

Logical handler ussd-expire-sessions:

1. Find active sessions where `last_seen_at` is older than configured timeout.
2. Mark `TIMED_OUT`.
3. Do not emit customer-facing notification for normal timeout.

This is a logical worker and can run in the channel app process.

11. Tests

1. New gateway session creates `ussd_session`.
2. Duplicate gateway request returns same response.
3. Unknown MSISDN shows safe fallback.
4. Balance flow calls billing API.
5. Payment instructions flow calls payment API but does not apply payment.
6. Ticket flow calls TCK API with idempotency.
7. Session timeout worker closes stale sessions.
8. CUST-INT logging is created for meaningful actions.